FLEET TEAM DIGEST

737MAX-FTD-34-22002

Issue Title : Momentary Erroneous Barometric (Baro) Altitude (ALT) Display Phenomenon

Airplane Model	ATA	Minor Model(s)
737MAX	3413-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/28/2022	03/27/2025	10/21/2024

Estimated Completion Date	Next Update Date
01/22/2026	06/17/2025

Revision Description

Updated Milestones Section.

Description

On the 737NG model aircraft, Boeing has received reports of momentary erroneous barometric-corrected altitude displayed on the Captain's Primary Flight Display (PFD), which was first reported in the year 2003. The Captain's altitude display briefly scrolls up or down and then returns to normal after a second or two. These events are infrequent, last a very short duration, and always self-recover. The root cause of this phenomenon has not been determined.

The 737NG and the 737 MAX share the same Air Data System (ADS) architecture and the altitude jump phenomenon has now been reported on the 737 MAX. Due to the short duration and self-recovery, the nature of this event is not judged to be confusing or create a hazard.

When the phenomenon was first discovered, the autopilot would prematurely enter Altitude Acquire Mode and pitch towards the briefly erroneous target barometric-corrected altitude. To prevent this response, the Flight Control Computer (FCC) software was modified to filter a brief erroneous baro-corrected altitude input. Reference 737NG-FTD-22-11002. The 737 MAX autopilot software contains the same prevention filter as the 737NG.

Boeing continues to search for the root cause of this phenomenon. Currently we are investigating the Air Data Inertial Reference Unit (ADIRU), the Display Processing Computer (DPC) and the Electronic Flight Instrument System (EFIS) Control Panels, all of which are involved in the calculation of barometric corrected altitude. The autopilot is protected against these altitude jumps and there are no other airplane effects other than the Captain's altimeter display.

Background

Boeing has had previous reports regarding the altitude jump phenomenon on the 737NG dating back to year 2003. The Captain's

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altitude display briefly scrolls up or down and then returns to normal after a second or two. The altitude jump phenomenon has been captured in flight data, and was once captured in a video. The behavior is believed to be due to single frame corruptions of the ADIRU's digital baro reference input data, leading the ADIRU to "correctly" compute baro-corrected altitude based upon an erroneous baro input. Where exactly the corruption is occurring (before or after being received by the ADIRU) has not been definitively determined.

To date, we have not been able to capture enough data on these events to determine a root cause. Troubleshooting these events yields no cause and results in No Fault Found (NFF) component removals. Generally, there are no fault codes or maintenance message recorded that are associated with such an event.

These events generally are only a visual on the display and have no airplane effects, although in earlier cases the autopilot system sometimes attempted to "chase" or "capture" the erroneous altitude. Since then, software changes were implemented in the autopilot to ignore short duration altitude excursions. Refer to 737NG-FTD-22-11002 article for further information.

These altitude variations have been captured in some cases by the Digital Flight Data Recorder (DFDR) in the baro correction output. Per design, the baro corrected altitude has four (4) duplicate sets of data generated from the same ADIRU ARINC bus output, each with a separate ARINC label. The Common Display System (CDS) provides two barometric correction inputs to each ADIRU. Each ADIRU outputs the baro corrected altitude #1 and #2 based on barometric correction input sources #1 (on-side CDS input) and #2 (off-side CDS input) respectively. The correlation of barometric correction input source to ARINC 429 output bus label for corrected altitude #3 and #4 can be selected and is based on the ADIRU Air Data Reference (ADR) pin program. Basically, if a left ADIRU, corrected altitude #3, output label 251, is based on barometric correction input source #1 and corrected altitude #4, output label 252, is based on barometric correction input source #2.

The baro corrected altitude parameters in the DFDR data are as follows:

Baro Cor Alt No 1 (label 204) is referenced to Baro Corr No 1 (label 235) which is the Captains Altitude display
Baro Cor Alt No 2 (label 220) is referenced to Baro Corr No 2 (label 237) which is the F/O's Altitude display
Baro Cor Alt No 3 (label 251) and Baro Cor Alt No 4 (label 252) are referenced to either Baro Cor Alt No 1 or 2 based on ADR pin program.

Review of these data labels in the recorded data, if the phenomena is captured, can help confirm if an erroneous altitude event on the Captain's side is the known altitude jump phenomenon and if further maintenance is needed.

Status

Boeing has worked with Honeywell to design a filter for eliminating those suspected single frame corruptions to mask the resulting jumping tape effect. An ADIRU software update, being developed for the 737-10 MAX sub-model, will contain this filter to prevent momentary erroneous baro correction data from causing a momentary erroneous altitude output.

Boeing also continue to collect reports and data from operators in support of an eventual root cause determination.

Boeing is updating the next revision of the Interactive Fault Isolation (IFIM) 34-20-00-810-805, Altitude Unreliable, to reflect the

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guidance provided in Interim Action below.

Interim Action

Boeing recommends Interactive Fault Isolation Manual (IFIM) 34-20-00-810-805, Altitude Unreliable, for a suspected altitude jump event occurring on the Captain's PFD. The IFIM is currently being updated to provide the following guidance for altitude jump events:

1) Check Display System BITE to ensure no disagree messages are captured.

2) Check ADIRU BITE history to ensure no air data system sensor input faults were logged during event.

3) Confirm no flight deck affects occurred other than the altitude display briefly scrolling up or down and then returning to normal after a second or two.

4) Static Port - Detailed Inspection TASK 34-11-02-200-801. If Foreign Object Debris (FOD) is found, then perform the Pitot Static System - Flushing TASK 34-11-00-170-802 only on the Captain static system.

NOTE: The ALT DISAGREE message should not be issued (the event must last over 5 seconds for the message to issue) if this is the Momentary Erroneous Barometric (Baro) Altitude (ALT) Display Phenomenon.

Monitor the airplane for repeat altitude unreliable issue. If repeated, perform further steps provided in Altitude Unreliable IFIM task.

Final Action

An ADIRU software update effort, in development for the 737-10 MAX, will contain this filter to prevent momentary erroneous baro correction data from causing a momentary erroneous altitude output. After certification on the -10, the update will be made available for all MAX models.

Milestones

1) 737MAX IFIM 34-20-00-810-805, Altitude Unreliable Update ECD: OCT 2024 (complete)

Operator Action

Contact Boeing via Boing Communication System (BCS) to report altitude jump event. The below information is requested:

1) FDR data

2) Flight crew event write-up description

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3) Maintenance actions and results

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Digest Article (FTD)	737NG-FTD-22-11002		

Related Categories

- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message
- Flight Operations : Flight Operations

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